A COMPLEX WITH A SINGLE NONVANISHING COHOMOLOGY GROUP SPLITS WITH THAT GROUP AS A DIRECT SUMMAND

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A complex whose cohomology vanishes outside one degree sits in a distinguished triangle with its truncations, and the triangle splits as soon as the connecting map is zero. invoked at proof lines 20807-20812. Produces the distinguished triangle and reduces the assertion to the vanishing of the connecting map.
- Over a regular local ring the Hom groups between the relevant truncations vanish in the degrees needed, which forces the connecting map to be zero. invoked at proof lines 20813-20817. Kills the connecting map by a degree count, which is where regularity enters.

Source reference. Original lemma 05QT: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 20811-20813 to reduce the statement to the vanishing of the third arrow; the degree computation showing that vanishing is inside the counted span.

Source reference. Original lemma 0FXJ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 20767–20767. Invoked at proof lines 20816-20817 for the vanishing of the Hom group between the truncations; the distinguished triangle and the reduction that set up that computation are inside the counted span.

Lemma 0.2. *Let R be a Noetherian regular ring of dimension $1 \leq d < \infty$. Let $K \in D(R)$ be perfect and let $k \in \mathbf{Z}$ such that $H^i(K) = 0$ for $i = k-d+2, \ldots, k$ (empty condition if $d = 1$). Then $K = \tau_{\leq k-d+1}K \oplus \tau_{\geq k+1}K$.*

Proof. The vanishing of cohomology shows that we have a distinguished triangle

$$\tau_{\leq k-d+1}K \to K \to \tau_{\geq k+1}K \to (\tau_{\leq k-d+1}K)[1]$$

By Derived Categories, Lemma 05QT it suffices to show that the third arrow is zero. Thus it suffices to show that $\mathrm{Hom}_{D(R)}(\tau_{\geq k+1}K, (\tau_{\leq k-d+1}K)[1]) = 0$ which follows from Lemma 0FXJ. □

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